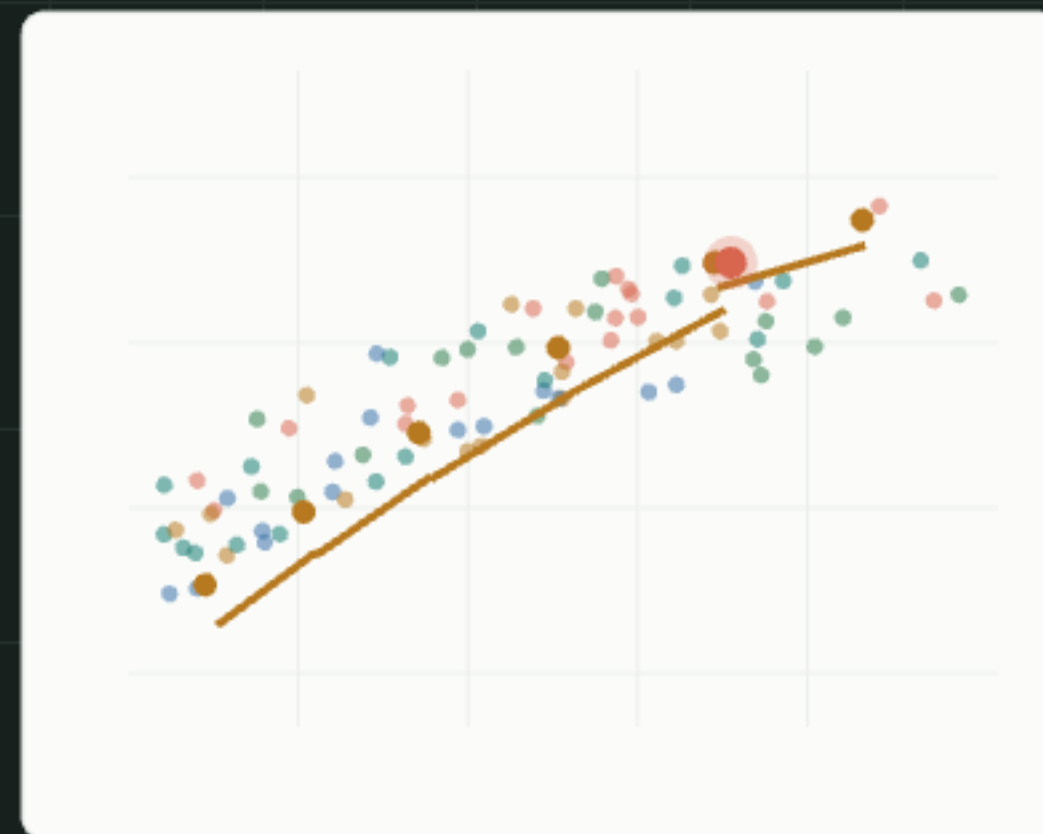


# A digital twin that lets members choose their best future self.

Counterfactual simulation turns longitudinal labs into an interactive planning surface, not another lab-result explainer.



future-space demo

Pareto frontier

**12,000**

simulated futures  
per member state

**180d**

forecast horizon  
biomarker movement

**48**

Pareto paths  
outcome vs burden

**retest**

feedback signal  
prediction error calibrates the twin

# Longitudinal panels change the AI job from interpretation to simulation.

The defensible capability is modeling biological motion: how biomarkers move together, and how future paths diverge under different choices.

## 1. Snapshot

Explain one panel: ApoB high, insulin elevated, hs-CRP noisy. Useful, but generic AI can do most of this.

generic

## 2. Trajectory

Repeated panels reveal slope, volatility, baseline drift, and cross-marker coupling. This starts to use Function's data advantage.

Function-specific

## 3. Future space

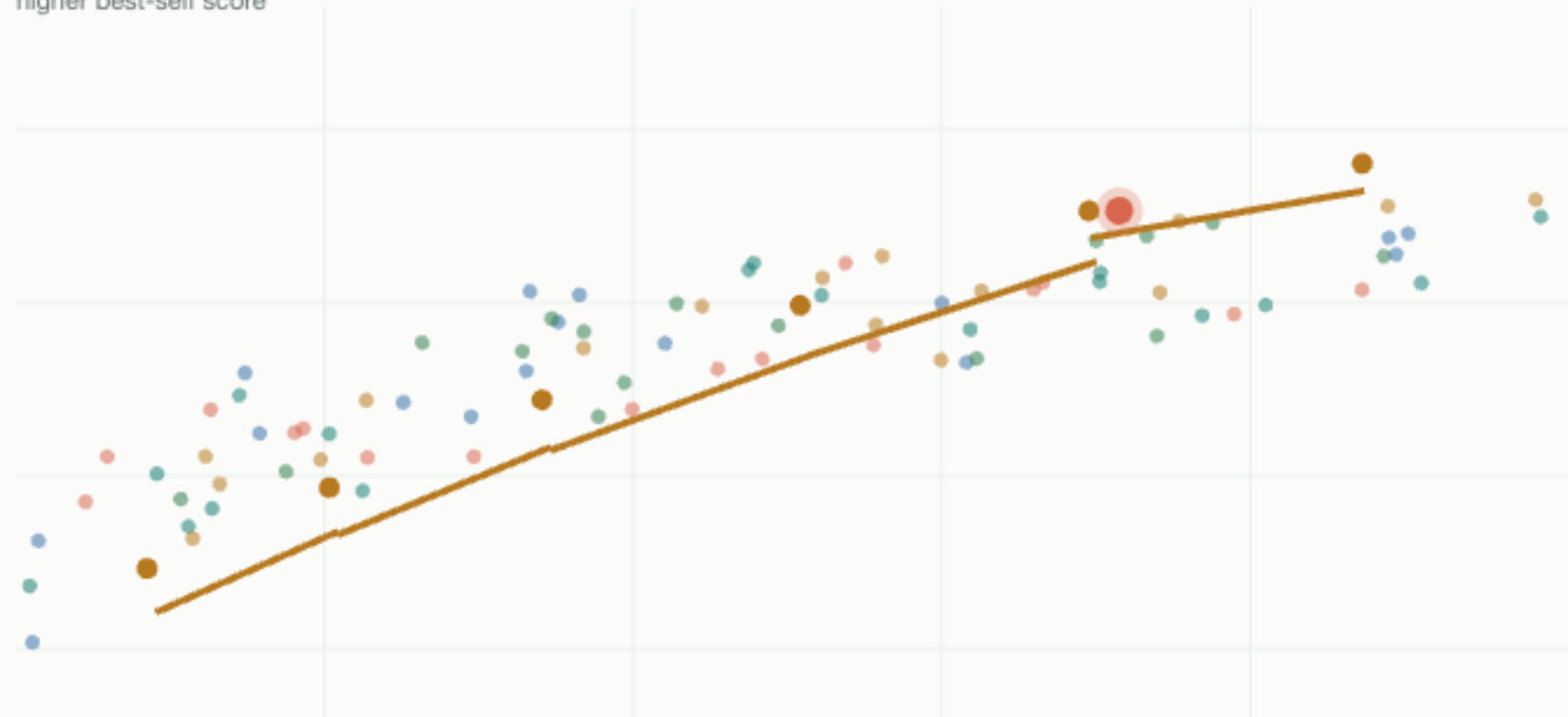
Simulate clinically plausible futures, expose tradeoffs, and let the member choose the path that matches their objective.

product magic

# Show the member a map of possible futures, then let them choose.

Each point is a simulated 180-day intervention path. The frontier makes tradeoffs explicit instead of hiding them behind one recommendation.

higher best-self score



intervention burden

## x-axis

intervention burden  
meds, effort, adherence load

## y-axis

best-self score  
forecasted biomarker improvement

## gold

Pareto frontier  
best outcome at each effort level

- Click any future to see projected ApoB, insulin, hs-CRP, ALT, and omega-3 curves.

# The interaction is values-first: users pick the future they can live with.

Walk the Head of AI through objective switches, then show how the forecast changes.

## Best overall

Balanced cardiometabolic improvement.

## Lowest burden

A smaller plan the member will actually follow.

## No-med path

Constraint-aware simulation, not moralizing.

## Fast ApoB drop

Outcome-specific optimization.

## Maximum change

Biggest movement, highest burden.

Say while clicking:

**“The twin does not claim there is one correct future. It exposes choices: speed, medication preference, effort, and confidence.”**

selected forecast

**ApoB -31**

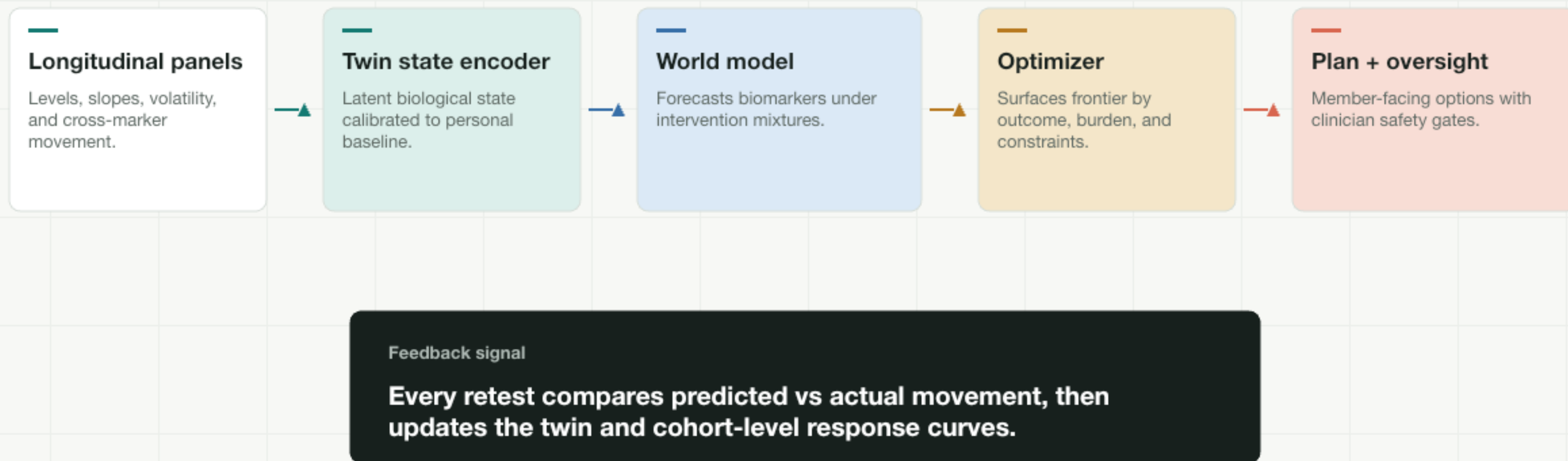
**Insulin -6.9**

**ALT -10**

Synthetic example: selected future vs no-change baseline.

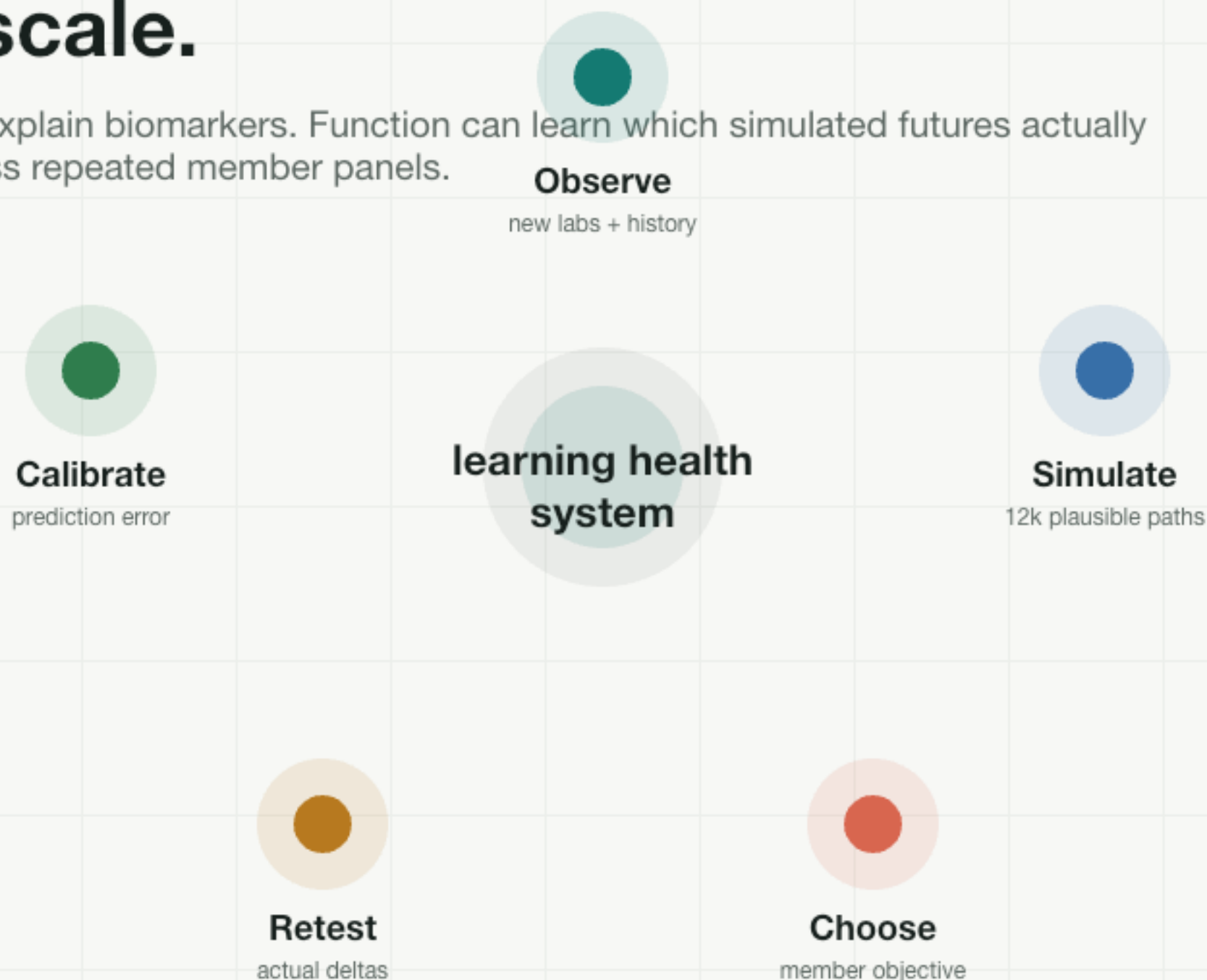
# Under the hood: a counterfactual world model plus multi-objective optimizer.

The architecture is narrow enough to validate: measurable biomarker dynamics, explicit uncertainty, and clinician-constrained action classes.



# The moat is not prediction. It is prediction error at retest scale.

Generic AI can explain biomarkers. Function can learn which simulated futures actually materialize across repeated member panels.



## closed loop

not a static model  
simulation improves after every panel

## personal

baseline-aware  
member-specific drift matters

## cohort

trajectory neighbors  
similar dynamics, not just demographics



# The credible version is constrained, measured, and clinician-aware.

Do not pitch autonomous medical advice. Pitch a biomarker simulator with validation gates and safe product boundaries.

1

## Retrospective backtest

Forecast held-out panels; measure calibration and directional accuracy.

2

## Off-policy evaluation

Estimate intervention effects while controlling for confounding and selection.

3

## Clinician review

Constrain action classes; surface contraindication and escalation checks.

4

## Prospective pilot

Run member-facing simulations with clinician-mediated recommendations.

5

## Production guardrails

Monitor drift, subgroup performance, uncertainty, and safety events.

Say this clearly

**“This is synthetic and not medical advice. The production system would be clinically constrained and prospectively validated.”**

# A chatbot explains your labs. A digital twin lets you rehearse your future.

Then every retest teaches Function whether that future was real.

Walkthrough move

**Show 12,000 futures,  
switch objectives, then  
move one what-if slider.**

Technical claim

**World model plus  
multi-objective  
optimizer on longitudinal  
state.**

Defensibility

**Every retest converts  
prediction error into a  
better member twin.**

**snapshot**

what happened

**trajectory**

what is changing

**future**

what could happen

**retest**

what became real